Lab 42 — Secure Data Sanitisation (Media Disposal)

The learner writes sensitive data to a virtual disk, then securely sanitises it and verifies the data is unrecoverable — the disposal stage of the asset lifecycle.

Run on https://killercoda.com/playgrounds/scenario/ubuntu

Objective: Securely wipe media and verify destruction

You'll build: A wiped virtual disk whose sensitive string is no longer recoverable, with before/after evidence.

Tools: dd, shred, grep, strings

Step 1 — Create a virtual disk with secret data

```
dd if=/dev/zero of=/disk.img bs=1M count=50 && echo "TOP-SECRET-DATA"
>> /disk.img
```

Step 2 — Confirm the secret is present

```
strings /disk.img | grep TOP-SECRET && echo FOUND
```

Step 3 — Securely overwrite the media

```
shred -v -n 3 -z /disk.img
```

Step 4 — Confirm the secret is gone

```
strings /disk.img | grep TOP-SECRET || echo "SANITISED — not recoverable"
```

Step 5 — Note the method for the disposal record

```
echo "Method: 3-pass overwrite + zero (NIST 800-88 clear)"
```

Verify

The secret string is found before shred and is absent after — sanitisation is verified.